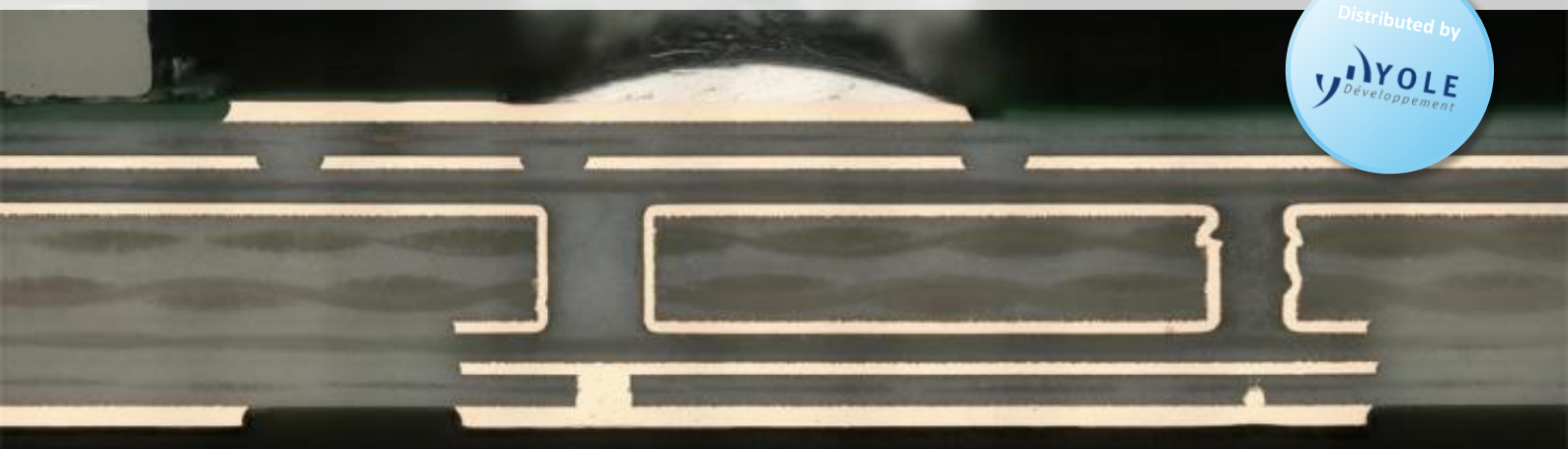
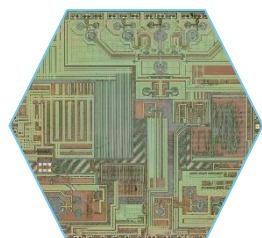
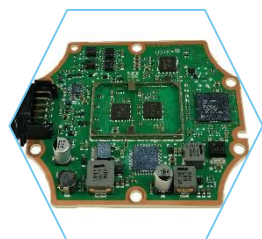


# **EXHIBIT 48**



## Denso DNSRR004 Short Range Radar

*A PTFE-free radar, with single receive-transmit MMIC based on TowerJazz's SiGe CMOS technology, for the first time associated with a phased array antenna.*



**Title:** Denso DNSRR004  
Blind Spot Radar

**Pages:** 76

**Date:** December 2019

**Format:**  
PDF & Excel file

Denso Corp has developed a 24GHz submillimeter-wave rear and side radar sensor for detecting vehicles in adjacent lanes. Used in several Toyota sedan models, the radar is embedded into the rear bumper of the vehicle and can detect following vehicles likely to enter the driver's rear and side blind spot.

The DNSRR004 includes a Monolithic Microwave Integrated Circuit (MMIC) manufactured using SiGe CMOS Technology from TowerJazz. It combines transmitter (Tx) and receiver (Rx) features on a single chip in a Quad Flat No-Lead (QFN) Package. To enable accurate detection whether moving forward or reversing, a phase shifter switches the sensing direction and range of the submillimeter-wave radar sensor. Radio wave transceiver and phase shifter functions run through separate integrated circuits to reduce the size of the sensor. The radio frequency printed circuit board substrate is completely PTFE free, and uses glass-reinforced hydrocarbon ceramic laminate and

integrates 20 Tx and 20 Rx patch antennas.

Based on a complete teardown analysis of the Denso radar, this report provides the bill-of-material (BOM) and the manufacturing cost of the radar sensor. The report also includes a physical analysis of the MMIC, complete cost analysis and selling price estimation.

### COMPLETE TEARDOWN WITH

- Detailed photos
- Block diagram
- RF chipset
- PCB cross-section
- Die picture and cost estimation of the MCU and MMIC
- Complete and priced bill-of-materials
- Manufacturing process flow
- Manufacturing cost analysis
- Estimated sales price

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**Nicolas Radufe** is in charge of physical analysis at System Plus Consulting. He has deep knowledge in chemical and physical analyses. He previously worked in microelectronics R&D for CEA/LETI in Grenoble and for STMicroelectronics in Crolles.

## RELATED REPORTS



### Aptiv R3TR 76GHz Short Range Radar

The first Aptiv short range radar with Infineon SiGe CMOS technology for Advanced Driver Assistance Systems (ADAS).  
December 2019



### Automotive Radar Comparison 2018

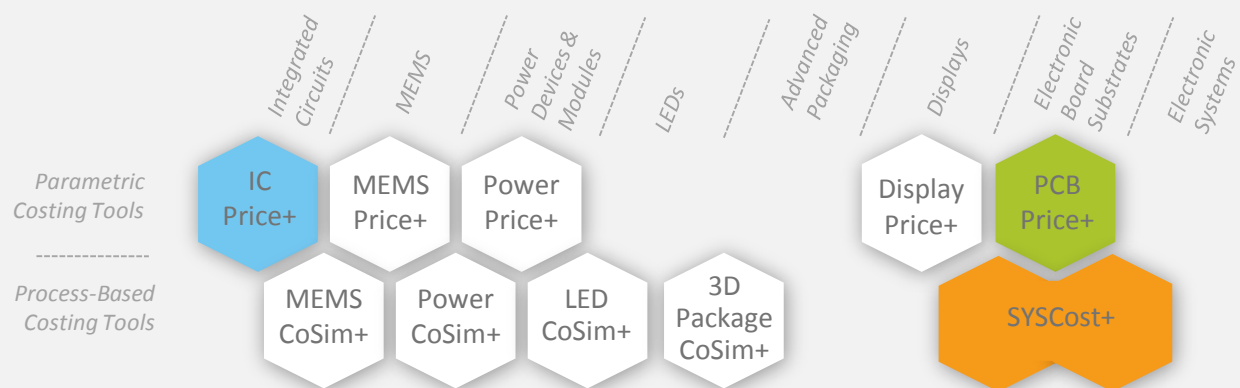
Continental, Veoneer, ZF, Valeo, Bosch, Aptiv, Denso, Ainstein: Discover the technologies used in the main Radar Systems and Chipsets.  
October 2018



### Aptiv's Third Generation of 77 GHz-Based Short-Range Radar (SRR3)

The first highly integrated 77GHz short range radar sensor for ADAS with a single receive-transmit MMIC bare die from Infineon found on the market.  
October 2018

## COSTING TOOLS



Our analysis is performed with our costing tools SYSCost+, IC Price+ and PCB Price+. System Plus Consulting offers powerful costing tools to evaluate the production cost and selling price from single chip to complex structures.

### SYSCost+

Provides all component costs estimation including PCB, housing and connectors, and a simulation of the assembly cost and test process at the board and system level.

### IC Price+

Performs the necessary cost simulation of any Integrated Circuit: ASICs, microcontrollers, memories, DSP, smartpower...

### PCB Price+

Performs the necessary cost simulation of a large type of PCB substrates according to different technical characteristics such as the material, the metal layers, the type of Vias, etc.

## WHAT IS A REVERSE COSTING®?

Reverse Costing® is the process of disassembling a device (or a system) in order to identify its technology and calculate its manufacturing cost, using in-house models and tools.



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    - within a reasonable time for Products ordered prior to their effective release. In this case, the Seller shall use its best endeavours to inform the Buyer of an indicative release date and the evolution of the work in progress.
  - 2.2 Some weeks prior to the release date the Seller can propose a pre-release discount to the BuyerThe Seller shall by no means be responsible for any delay in respect of article 2.2 above, and including incases where a new event or access to new contradictory information would require for the analyst extra time to compute or compare the data in order to enable the Seller to deliver a high quality Products.
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To ensure the payments, the Seller reserves the right to request down payments from the Buyer. In this case, the need of down payments will be mentioned on the order.

3.4 Payment is due by the Buyer to the Seller within 30 days from invoice date, except in the case of a particular written agreement. If the Buyer fails to pay within this time and fails to contact the Seller, the latter shall be entitled to invoice interest in arrears based on the annual rate Refi of the «BCE» + 7 points, in accordance with article L. 441-6 of the French Commercial Code. Our publications (report, database, tool...) are delivered only after reception of the payment.

3.5 In the event of termination of the contract, or of misconduct, during the contract, the Seller will have the right to invoice at the stage in progress, and to take legal action for damages.

## 4. Liabilities

- 4.1 The Buyer or any other individual or legal person acting on its behalf, being a business user buying the Products for its business activities, shall be solely responsible for choosing the Products and for the use and interpretations he makes of the documents it purchases, of the results he obtains, and of the advice and acts it deduces thereof.
- 4.2 The Seller shall only be liable for (i) direct and (ii) foreseeable pecuniary loss, caused by the Products or arising from a material breach of this agreement
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- 4.5 All the Products that the Seller sells may, upon prior notice to the Buyer from time to time be modified by or substituted with similar Products meeting the needs of the Buyer. This modification shall not lead to the liability of the Seller, provided that the Seller ensures the substituted Product is similar to the Product initially ordered.
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- 4.7 The deadlines that the Seller is asked to state for the mailing of the Products are given for information only and are not guaranteed. If such deadlines are not met, it shall not lead to any damages or cancellation of the orders, except for non acceptable delays exceeding [4] months from the stated deadline, without information from the Seller. In such case only, the Buyer shall be entitled to ask for a reimbursement of its first down payment to the exclusion of any further damages.
- 4.8 The Seller does not make any warranties, express or implied, including, without limitation, those of sale ability and fitness for a particular purpose, with respect to the Products. Although the Seller shall take reasonable steps to screen Products for infection of viruses, worms, Trojan horses or other codes containing contaminating or destructive properties before making the Products available, the Seller cannot guarantee that any Product will be free from infection.
5. Force majeure

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6. Protection of the Seller's IPR
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8. Miscellaneous

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The Seller may, from time to time, update these Terms and Conditions and the Buyer, is deemed to have accepted the latest version of these terms and conditions, provided they have been communicated to him in due time.

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- 9.2 French law shall govern the relation between the Buyer and the Seller, in accordance with these Terms and Conditions.

